

Supplementary Material

Effects of arrival timing, source startup and room reflections on three-microphone sound localization

Aneesh Arnav Chikkala

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This document explains the simulation, numerical checks and additional results in plain language. The study asks how three choices affect a compact microphone array: whether arrivals are placed between samples or rounded to a sample, whether the source has just started or was already running, and whether room reflections are present. All numbers below come from the corrected final simulation campaign.

This is a controlled simulation study. It does not claim that the same errors will occur with physical microphones, speech, moving sources or every room.

1 Array and direction estimate

Three microphones sit at the corners of an equilateral triangle. Each microphone is 5 cm from the centre, so neighbouring microphones are 8.66 cm apart. The source is 1.5 m from the centre and at the same height. Sound speed is 343 m/s and the sampling rate is 48 kHz. The microphone coordinates are

$$\mathbf{m}_i = R[\cos \phi_i, \sin \phi_i]^T, \quad R = 0.05 \text{ m}. \quad (1)$$

The arrival time at a microphone is its distance from the source divided by sound speed. Pair delays are differences between those arrival times:

$$t_i = \frac{\|\mathbf{s} - \mathbf{m}_i\|}{c}, \quad \tau_{ij} = t_i - t_j. \quad (2)$$

GCC-PHAT estimates the three pair delays. A least-squares plane-wave solve converts them into a direction vector, and the reported azimuth is the angle of that vector:

$$\tilde{\mathbf{u}} = -cA^+ \hat{\boldsymbol{\tau}}, \quad \hat{\theta} = \text{atan2}(\tilde{u}_y, \tilde{u}_x). \quad (3)$$

Because the source is at a finite distance, its wavefront is slightly curved. This creates a small model mismatch even when the direct arrival times are exact.

2 Room and signal model

Each room is a rectangular enclosure with mirror-image reflections. Wall reflection coefficients do not vary with frequency. The model includes distance spreading, but not diffraction, diffuse scattering, air absorption or microphone directionality. A reflected path of length r has pressure gain

$$g(r) = \frac{1}{4\pi r} \prod_w \beta_w^{N_w}. \quad (4)$$

The corrected model retains paths up to 1.2 s, uses image extent 76 and constructs fractional arrivals on a grid 32 times finer than the 48 kHz recording grid. A Kaiser-windowed sinc filter converts that response to the acquisition rate. The comparison condition rounds each path to its nearest acquisition sample while leaving its gain and later processing unchanged.

The source is Gaussian noise restricted to 300–3400 Hz. Every microphone uses the same causal Butterworth bandpass filter. Added noise is scaled relative to the filtered direct signal:

$$\text{SNR} = 10 \log_{10} \left(\frac{P_{\text{direct}}}{P_{\text{noise}}} \right). \quad (5)$$

A steady recording includes enough source history for the room response and filter transient to settle. In an onset recording, the source is silent before analysis begins. The analysed source samples and microphone-noise samples are otherwise identical.

3 Study design

The study uses rooms measuring $6 \times 5 \times 3$ m, $5 \times 4 \times 2.8$ m and $8 \times 6 \times 3.2$ m. Each room has a centred and an offset array position, three nominal reverberation settings (0.15, 0.30 and 0.60 s), and 12 source directions. This gives 216 physical scenes.

The main study evaluates 0, 10 and 20 dB SNR with five independent recordings per scene. Each recording contains 32 non-overlapping frames of 2048 samples, or 1.365 s. A separate matched study uses 72 scenes at 10 dB and tests all eight combinations of fractional or rounded arrivals, steady or onset history, and direct or reflected propagation.

Temporal averaging uses separate 128-frame recordings. Five recordings per scene estimate the averaging model and five different recordings test it. This prevents the model from being judged on the same random signals used to fit it.

Table 1: Numbers of frame-level estimates in the final campaign.

Part	Scenes	Estimates
Main room study	216	103,680
Seven matched variants	72	80,640
Long recordings	72	92,160
Total		276,480

4 Error and uncertainty

Angular error is wrapped from -180° to 180° , so crossing the direction boundary does not create a false large error. The main summary is root mean square error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N e_n^2}. \quad (6)$$

RMSE gives extra weight to large mistakes. The analysis also records median absolute error, the 90th percentile, circular bias and the share of estimates with more than 5° absolute error.

Uncertainty intervals use 2,000 bootstrap draws. Whole recordings are resampled within each fixed scene, and matched conditions receive the same resampling. Frames from one recording are never treated as separate experiments. These intervals describe noise variation in the tested scenes, not every possible room.

Directions are averaged with a circular mean so values near -180° and 180° remain close. The covariance model is interpreted only for scenes that met preselected concentration rules. Every scene remains in the empirical held-out results.

5 Numerical checks

The first campaign used a 0.9 s response and image extent 48. A stronger joint check showed that ten of 960 estimates changed by more than the 0.05° numerical target, with a largest change of 0.16551° . Those results were excluded.

The final campaign uses a 1.2 s response, image extent 76 and propagation factor 32. Extending the model to 1.5 s and extent 96 changed the tested estimates by at most 0.01273° . An independent exact-phase room calculation differed by 0.000807° for the selected reference scene. Direct-path, interpolation and angular-grid checks also remained below 0.05° .

These are numerical convergence checks, not proof that the simulated acoustics match a real room. The complete final results remain separate from the rejected earlier campaign.

6 Main results

Table 2: Main findings at 10 dB SNR.

Question	Result
Does sample rounding matter?	Direct-path RMSE rose from 0.923° to 2.300° .
Do reflections matter?	With fractional steady signals, reflected propagation added 747.97 deg^2 of matched mean squared error.
Does startup history matter?	In the high-reflection subset, first-frame RMSE was 4.157° at onset and 42.508° after the reflected field had built up.
Does averaging help?	Across 128 frames (5.46 s), held-out RMSE fell to 0.835° in the low subset and 10.158° in the high subset.

The low and high subsets change both array placement and wall absorption, so their difference represents a combined acoustic condition. The onset advantage is temporary: by the final frame of the 1.365 s recording, onset and steady conditions both reached 41.526° RMSE in the high subset.

Rounding, reflections and startup do not behave as independent errors. Reflections reshape the correlation function, so a timing-grid change can change which peak GCC-PHAT selects. A matched factorial comparison is therefore more informative than subtracting two headline RMSE values.

Increasing SNR from 10 to 20 dB helped the least reflective condition but barely changed the larger errors in the more reflective settings. Reflected structure, rather than microphone noise alone, dominated many difficult scenes.

7 Timing-precision check

A separate analytical bound asks how far the fitted direction can move if each microphone arrival time has an error no larger than h . It keeps the dependence between microphone pairs because the same microphone appears in more than one pair. For an equilateral array,

$$\rho = \frac{4ch}{3R}. \quad (7)$$

A sufficient condition for an angular change no greater than η is

$$h \leq \frac{3R}{4c} \sin \eta. \quad (8)$$

For $R = 0.05 \text{ m}$, $c = 343 \text{ m/s}$ and $\eta = 0.5^\circ$, the sufficient limit is $h \leq 0.954 \text{ } \mu\text{s}$ per microphone arrival. This is a conservative check for ideal arrivals. It is not a required hardware sampling interval and does not predict GCC-PHAT errors in noise or reverberation. All 34,560 rounded-arrival cases and 12,000 additional random perturbations stayed within their calculated bounds.

8 Sensitivity and second estimator

A fixed 24-scene subset tested a 17° array rotation, an unequal triangle, unequal wall absorption and a narrower 300–1700 Hz source. These used new random recordings, so their differences are sensitivity observations rather than paired causal effects.

SRP-PHAT was evaluated on the same base recordings as GCC-PHAT. It had higher RMSE (30.262° versus 22.413°) but slightly fewer estimates above 5° (28.62% versus 30.57%). The summaries emphasize different parts of the error distribution, so this restricted comparison does not establish that either method is generally better.

9 Interpretation and limits

The main practical result is that simulation setup changes the apparent difficulty of localization. Nearest-sample arrivals add avoidable error. Starting the recording at the same moment as the source temporarily removes reflections from earlier emissions and can make the first frame look unrealistically easy. Averaging helps, but it cannot reliably remove persistent bias or repeated selection of the wrong correlation peak.

The study uses one stationary synthetic source, ideal matched microphones, three rectangular rooms and horizontal direction only. Reflections are specular and frequency independent. The model excludes microphone mismatch, clock drift, calibration errors, directivity, diffuse scattering, moving sources, multiple sources, speech and real hardware. The results therefore describe the stated simulation and are not measured device accuracy.

The bootstrap intervals condition on the selected scenes. They do not show how the method would vary across all real rooms. A measured-room experiment is required before making hardware or real-world performance claims.

10 Reproduction record

The final campaign retains the frozen configuration, deterministic seed identities, complete frame-level outputs, scene metadata, analysis summaries and numerical checks. A clean-environment representative rerun reproduced 8,640 primary estimates and all six main figures. This was a representative check, not a second complete run.

No raw microphone recording is required because the reported experiment is synthetic. The code and retained outputs should be deposited in a stable research repository if they are shared publicly. The repository record should identify the release version and include the dependency lock file. The permanent repository URL or DOI should be added to the paper before submission, and suitable software and data licenses should be selected before public release.